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Leveraging Multi-Well Pad Development to Maximize Value in a Deep Tight Gas Condensate Field: A Groundbreaking First in Oman

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Abstract

This paper presents the first full-scale implementation of multi-well pad drilling in Oman's deep tight gas reservoirs, specifically within Block 10, targeting the Miqrat and Barik formations at depths of 4500–5000 meters. To address the challenges of low permeability, complex completions, and remote desert operations, the team adopted a multi-well pad strategy using S-shaped deviated wells and a walking rig, enabling efficient batch drilling. Each pad hosts three wells spaced 10 meters apart, matured in integrated clusters that streamlined planning and cut the maturation timeline by nearly one-third. Pad locations were optimized using high-resolution geological models to ensure effective reservoir drainage and minimal surface impact, while a 4-string slim well design with directional profiles up to 33° and step-outs of 1400 meters was employed. Completion and testing operations utilized a stack-frac approach with an Integrated Frac Services fleet, achieving significant reductions in mobilization time and enabling green completions with up to 90% GHG emission reduction. Operational synergies included shared BHAs, mud systems, and real-time benchmarking across wells. This project sets a new technical benchmark for tight gas development in Oman, demonstrating how integrated planning, pad optimization, and collaborative execution can deliver scalable, cost-effective solutions in complex onshore environments.

Introduction

The Sultanate of Oman is actively expanding its domestic gas production to meet growing energy demands and support long-term economic diversification. A key component of this strategy is the development of deep, tight gas condensate reservoirs in the interior of the country, where geological complexity, remote terrain, and infrastructure limitations present significant technical and operational challenges. Among the most ambitious of these developments is the Block 10 gas fields. This project targets Miqrat and Barik tight sandstone reservoirs at depths of 4500–5000 meters (Al Hajri et al., 2023).

Tight gas development in Oman is inherently complex due to the low permeability of the reservoirs, the need for multistage hydraulic fracturing and the logistical burden of operating in remote desert environments (Lukmanov et al, 2018). Traditional single-well development models often lead to high well delivery costs, extended timelines, and fragmented infrastructure. To overcome these limitations and unlock the full

potential of the field, the project team introduced a multi-well pad development strategy; a first-of-its-kind implementation in Oman.

The Block 10 field is developed using a combination of single-well pads and multi-well pads. The Phase 1 wells have been drilled vertically from single-well pads. These wells were drilled in a 4-string drilling design. The Phase 2 wells, however, consist of 33 wells where they will be drilled from 11 multi-well pads (3 wells in each Pad). These wells are S-shaped deviated wells with a step-out between 1000-2000 m, depending on subsurface targets. In addition, these wells are only 10m apart from surface location and are batch-drilled using a walking rig to increase efficiency and lower well cost. Figure 1 illustrates the difference between single and triple-well pad and also shows the step-out of each well from surface location.



Figure 1—Block 10 tight gas field in the central part of Sultanate of Oman

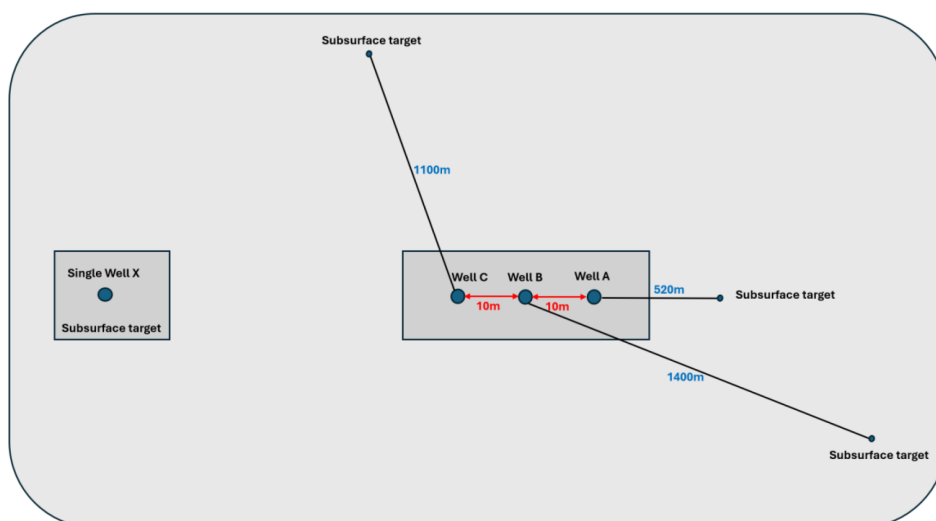


Figure 2—Illustration of Single well surface location and Triple-well pad where wells are 10m apart and drilled as S-shape wells to reach required step-out

The triple-well pad layout is such that it allows the extension of the pad by a second cluster of three wells. This second cluster will be sufficiently spaced from the first cluster so that the second cluster will be drilled without stopping the production of the three wells of the first cluster.

The objective of this paper is to present the planning, execution, and outcomes of this innovative development strategy. It highlights how the multi-well pad approach improved drilling efficiency, reduced non-productive time, and enhanced cost-effectiveness while maintaining high safety and environmental standards. As the first large-scale application of this model in Oman, the project sets a new benchmark for tight gas development in the region. The insights and lessons shared in this paper aim to inform future developments in similar geological and operational contexts, offering a scalable and repeatable model for efficient resource extraction in challenging onshore environments.

Strategic Development Objectives

The strategic implementation of a multi-well pad development approach in Block 10 was driven by the need to unlock the full potential of Oman's deep tight gas resources while maintaining cost discipline and operational efficiency. The primary objective was to maximize project value by reducing well delivery costs, accelerating time to production, and enabling scalable development across the lifecycle of Block 10 (Al Kindi et al, 2025).

Tight gas reservoirs such as the Miqrat and Barik formations present significant challenges due to their low permeability, deep burial depths, and the need for complex completions. Traditional single-well development models are not only capital-intensive but also logistically inefficient in remote desert environments. To address these constraints, the project team was tasked with identifying a well-pad development concept that would minimize land footprint, reduce development and operational costs while ensuring high drilling efficiency and HSE performance. Hence adopted a multi-well pad strategy for Block 10 development.

The multi-well pad strategy was designed to deliver several key benefits. It enabled operational efficiency through batch drilling, standardized well designs, reduced rig mobilization and reduced Frac and test equipment during Fracturing operations. Flowlines for multi-well pad will be pre-installed to the side of the location to enable Completion and Well Intervention (CWI) to frac and test directly into the pipeline without flaring gas. These wells will also share surface infrastructure such as remote manifold stations (RMS), bulk lines, and centralized export pipelines. While each pad has a larger surface footprint, the overall environmental and logistical impact is reduced by minimizing the number of access roads, utility corridors, and standalone facilities required across the field.

Pad Design and Well Placement

The design of multi-well pads and the placement of wells in the Block 10 field was guided by a combination of technical, geological, and logistical criteria aimed at maximizing reservoir coverage, minimizing surface impact, and enabling efficient field development. Given the depth and tight nature of the Miqrat and Barik sandstone reservoirs, precise well placement and pad optimization were essential to ensure effective reservoir drainage and long-term production performance.

The pad design process began with the integration of high-resolution geological and reservoir models. These models incorporated structural interpretations, petrophysical data, and dynamic simulation outputs to define optimal well trajectories and drainage patterns. The Miqrat reservoir, which accounts for approximately 85% of the estimated ultimate recovery (EUR), was prioritized in the modeling process due to its greater contribution to field deliverability. The Barik reservoir, while contributing a smaller portion of the EUR, was also considered in the vertical stacking of targets to maximize resource recovery from each wellbore.

Centralized pad locations were selected based on a combination of subsurface and surface criteria. Subsurface considerations included reservoir quality, structural continuity, and the ability to drill multiple wells with minimal deviation while maintaining target accuracy. Surface criteria included topography, accessibility, proximity to existing infrastructure, and environmental sensitivity. The goal was to minimize the number of pads required while ensuring full reservoir coverage, which led to the adoption of a 2 km² well spacing density.

Directional drilling techniques were employed to reach multiple reservoir targets from a single pad, reducing the need for additional surface locations. Batch drilling of similar well sections across pads further enhanced efficiency by allowing the reuse of bottomhole assemblies, drilling fluids, and operational workflows.

Surface infrastructure was consolidated around each pad to reduce environmental and logistical impact. Remote manifold stations (RMS) were strategically positioned to collect production from multiple pads and route it through bulk lines to a central export manifold station (CEMS). This design minimized the need for extensive surface pipelines and reduced the overall footprint of the development. Additional infrastructure, including access roads, utilities, and water management systems, was co-located with pad clusters to further streamline operations and reduce land disturbance.

The pad design also accounted for future scalability. In Block 10 Phase 3, the development plan includes the flexibility to extend existing pads or add new ones based on reservoir performance and production trends. This approach ensures that the field can be adapted over time without requiring major redesigns.

Integrated Planning and Collaboration

The first trial of multi-well pad drilling in Oman represents a significant milestone in the country's upstream operations, introducing a more efficient and collaborative approach to well maturation and execution. Given the novelty of this method in the region, it was crucial to establish strong connections with other Shell operations globally, particularly in Argentina and Canada, where multi-well pad drilling has been successfully implemented. These collaborations provided valuable insights into best practices, challenges, and solutions related to the maturation process and cross-departmental execution.

In this initial trial, the wells were selected and matured in a single integrated session, rather than through three separate sessions (as used in single well picking session). This approach allowed the surface location team to conduct a comprehensive survey for all three wells in one visit, significantly reducing logistical complexity and time. The wells were then matured in clusters of three per pad, with each well proposal developed in parallel. The surface locations of the wells were spaced approximately 10 meters apart, and each well was designed to follow an S-shaped trajectory—diverging below a certain depth and returning to vertical above the primary reservoir target. This design ensured optimal reservoir coverage while maintaining surface efficiency.

One of the key achievements in this process was the consolidation of the technical assurance sessions by combining the assurance for all three wells into a single session where the technical team was able to streamline the review and approval process. This not only reduced the time required for technical assurance but also minimized the workload for technical team, who benefited from the geological, geohazard, and fluid fill similarities among the wells to fill in their well proposals. The shared characteristics of the wells allowed for a more cohesive and efficient evaluation, eliminating redundant efforts and documentation.

As a result of these optimizations, the team achieved a substantial reduction in the overall maturation timeline—saving nearly one-third of the time typically required in single well maturation. This improvement aligns well with previously defined key performance indicators (KPIs) for well delivery and demonstrates the potential of multi-well pad drilling to enhance operational efficiency in Oman. The success of this trial also highlights the importance of global collaboration within Shell, showcasing how knowledge sharing and coordinated execution can lead to tangible performance gains.

The demonstrated benefits in time savings, resource utilization, and cross-functional coordination provide a compelling case for broader adoption of this approach.

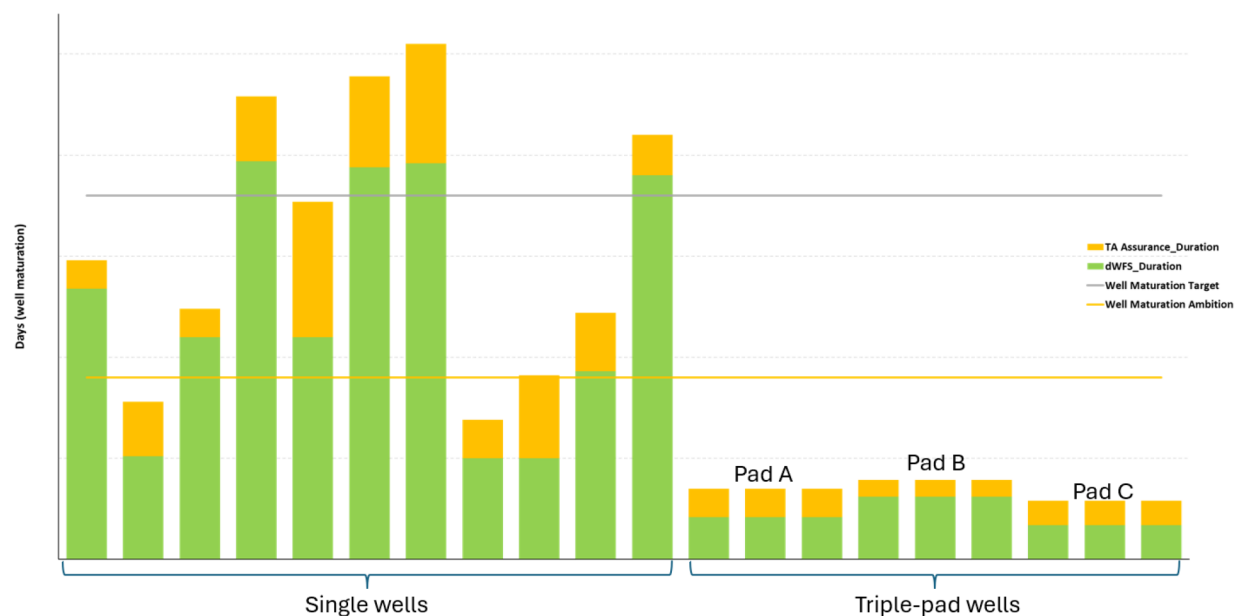


Figure 3—Maturation period for single wells vs. multi-well pad where timing saving was observed during well proposal maturation

Drilling Execution and Batch Operations

The rig has been upgraded to be as a walking rig that can skid easily from one well to another. The wells in each pad are spaced 10m away from each other where the main parts of the rig like the generators, mud tanks and mud pumps are not moved while the rest of the rig can move on a sliding rail (Kam et al, 2014). The time to skid from well to well is around 9.5 hrs on average as shown in the below graph.

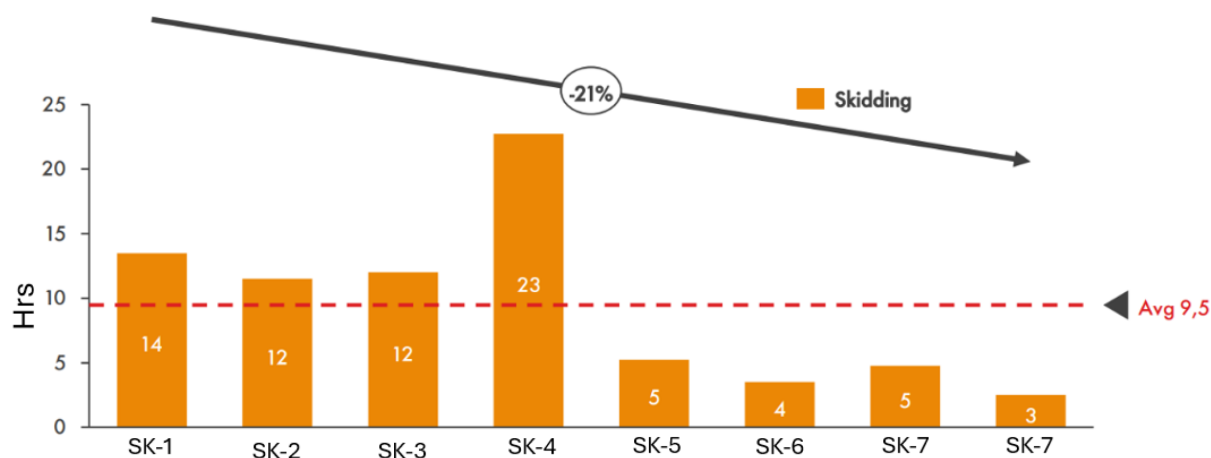


Figure 4—Skidding (SK) from each well to another within the same pad for different sections

The Pad A project represents the first successful implementation of multi-wall pad drilling in Phase 2 of Block 10 in Shell Development Oman (Deep Gas Development Wells), delivering a step change in simplification and efficiency. Three wells (Well A, B and C) were drilled consecutively from a single pad to reach down hole targets that are 500– 1,400 m apart. The wells started 10m apart for the first 200m then started to nudge Wells B and C to maintain a 40m distance from each other while drilling Well A as

a straight vertical as shown in the below figure. The three wells then continued to be vertical until passing the troublesome losses zone in the third section of the wells around 2000m then each well was directionally drilled in a different direction to meet the subsurface reservoir target.

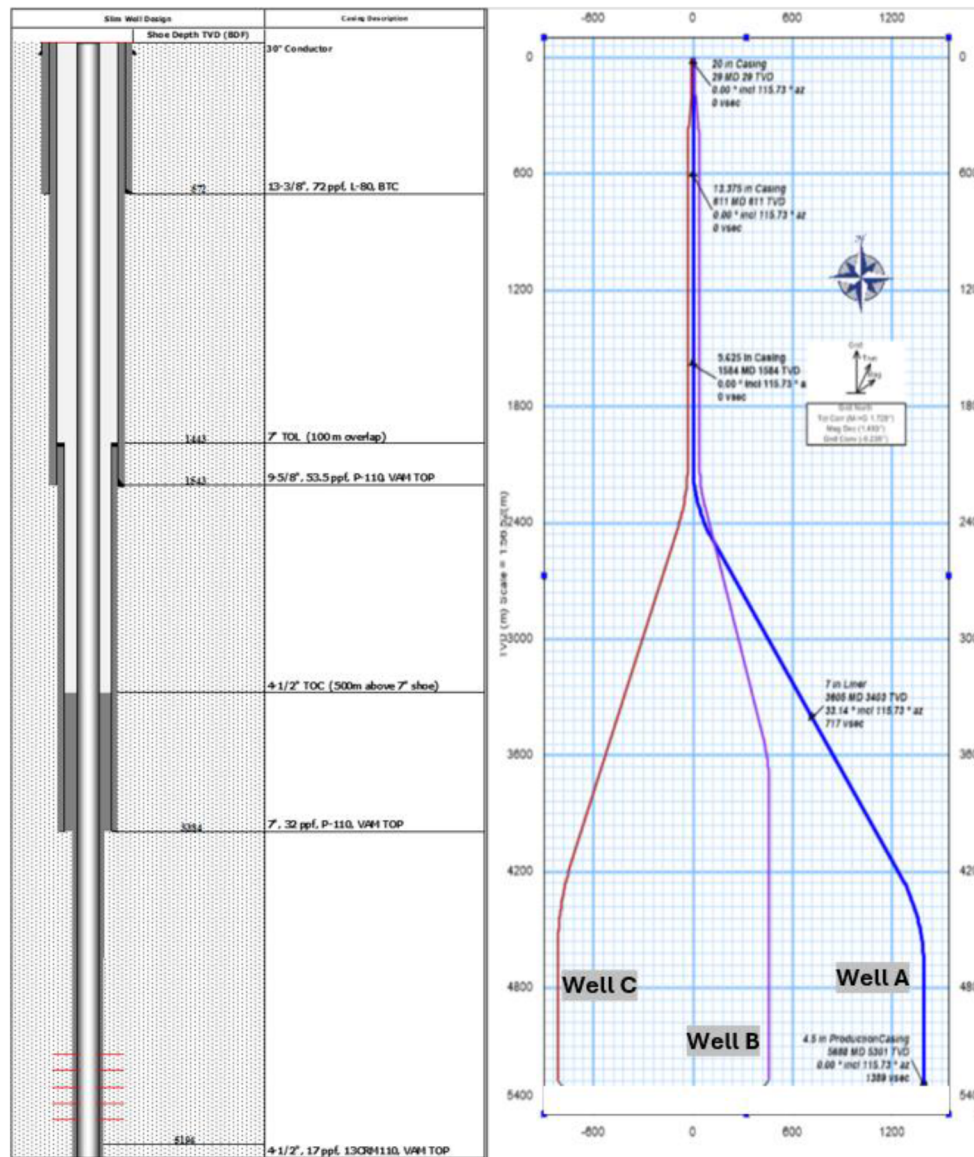


Figure 5—Cross sectional view of the first triple-pad wells

The well design for these 3 wells is a 4-string slim design and they were drilled by batching every section as below figure illustrates. The drilling starts with drilling and casing the first section of the three wells then starting drilling the next section where the last section was completed.

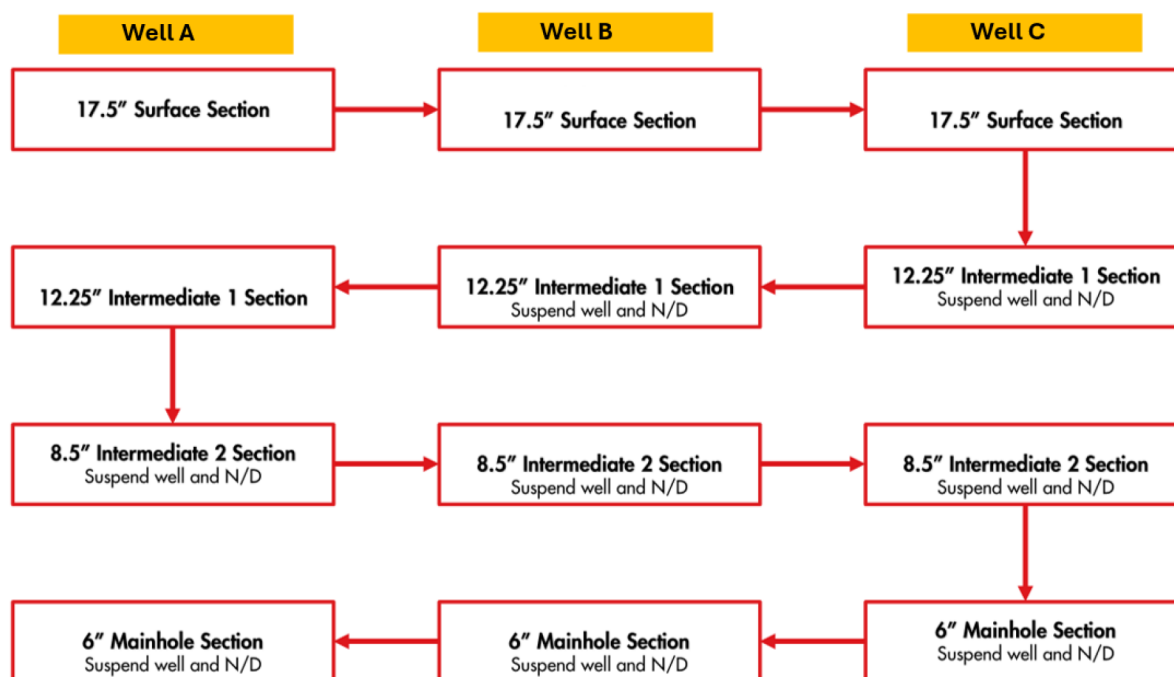


Figure 6—Batch drilling each section of the three wells then moving down to next section

This campaign achieved Goal Zero safety (No HSE Incidents). Directional plans were delivered to Total Depth in all wells, despite directional profiles with 33° tangent and 1,400m total step-out. Production hole sections were drilled matching AFE targets and open-hole wireline logging objectives were achieved across all three wells as per plan.

Below are some of the synergies achieved from the multi well pad wells drilling:

- Multi-well pads enabled a single drilling program and decision gate process, reducing repetition and improving efficiency across the campaign.
- Reduced Rig Move Time & Cost.
- Saving the cost of constructing 3 single well pads with their respective water and waste pits.
- Using shared Bottom Hole Assembly (BHA) and drill pipes (DP) between all 3 wells. As batching is done for each section, BHA /DP were racked back to drill all 3 wells with the same configuration reducing picking up and lay down operations in time and safety aspects.
- Shared mud systems, significantly reducing operational waste, logistics complexity and cost. Oil Based Mud (OBM) is used in the second section of the wells and hence a great cost and time has been saved in dealing with it only once for the three wells.
- Replicating effective drilling roadmaps for the section in following wells of the pad, enabling real-time benchmarking between wells and reducing trial and errors.
- Losses continued being the main challenge while drilling, however with the nature of the pad drilling all learnings from the first well were used to refine the plan of second and third wells allowing faster learning curve and effective execution.

As this is the first multi-well pad to be drilled, there were some challenges were faced and further optimizations are planned for future pads based on the learning from the first pad:

- Due to the deviated profile which resulted in drilling Gharif formation (Problematic reactive shale) with higher inclinations, this caused some challenges in running 7" liner across Gharif formation,

a geo mechanical study will be conducted to understand the stresses and the Mud weight (MW) requirements to ease tripping / running Liner and further optimize drilling operations.

- Discussions with different stake holders about the max inclinations allowed to enter reservoirs can also help to optimize the directional plan which will lead to decreasing inclinations in problematic Formations.

Hydraulic Fracturing and Testing Integration

Hydraulic Fracturing and Testing on a pad with 3 wells presented many opportunities for the Completion & Well Interventions (CWI) team especially that it is being trialed for the first time in Oman. Preliminary reviews drew on Shell's experience globally in high frac intensity basins and through our Value Stream Mapping (VSM) exercise the team captured key value drivers that can reduce critical path time.

It was identified that due to constraints with proppant transport and loading and manifold rig up time and cost, we could not realize critical path improvement with the application of frac tree manifolds and zipper frac process (Ogoke et al, 2014). As a result, a 'stack-frac' approach was selected instead of a 'zipper frac' one. Essentially, operations would entail perf then frac all the zones in one well before moving to the next well (see Fig.7), thus swapping between these two operations on the same well. Whereas zipper frac on the other hand would entail wireline perf on all three wells then frac on all 3 wells and repeat. Ultimately, the key components selected for implementation on multi well pads were additional water and proppant storage, flowback/well test manifold, 200T crane for single lift of Pressure Control Equipment (PCE) and Green Completions (Flow to Host).

Pad A: FRAC & Test scope

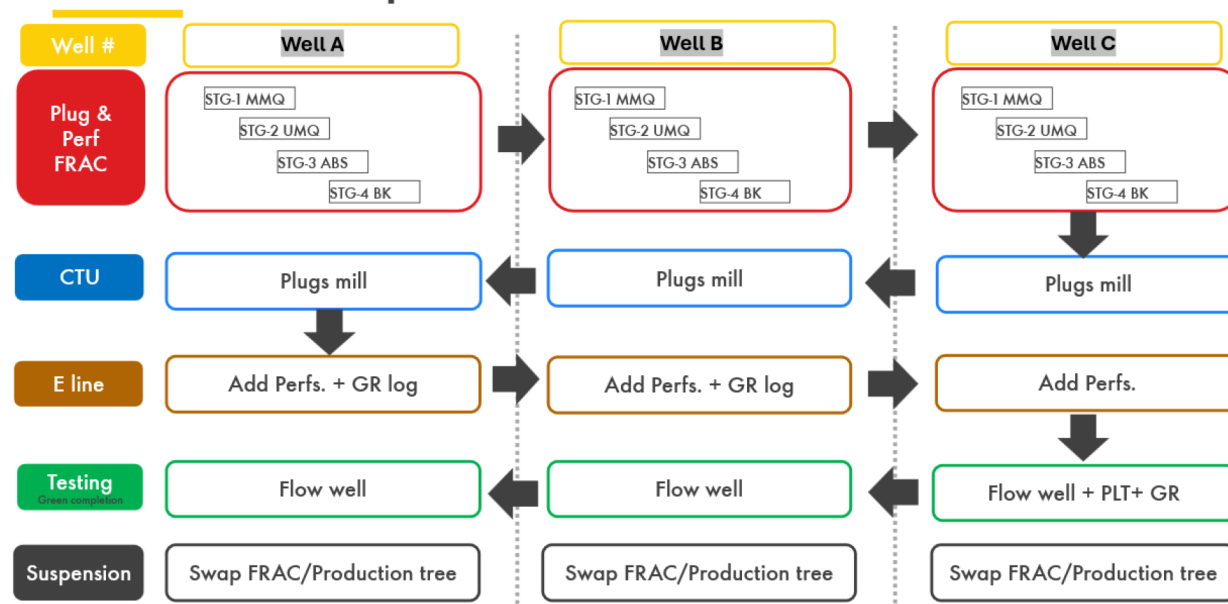


Figure 7—Frac and Test operation sequence for the first multi-well pad

As we got closer to the start of the campaign the CWI team along with our service providers completed several team engagements starting with Performance Improvement Identification and culminating in Complete Well on Paper (CWOP) exercise to ensure operational delivery of these initiatives.

It's worth noting, Shell Development Oman (SDO) deploys an Integrated Frac Services (IFS) model to stimulate their wells, meaning frac pumping, slickline, wireline and coil tubing are all mobilized together to the wellpad. This allows us to cater to any operational requirements without interruption to the campaign, e.g. screenout, refrac, etc. One of the main challenges with this approach is the mobilization and rig up

time, and so a key value driver for the multi well pad approach is the shared mobilization time. Instead of performing the move 3 times, one mobilization time is shared between the 3 wells; a 50% reduction in preparation phase compared to our single wells.

Furthermore, the well test package is mobilized from the start and manifolded to all three wells. Not only did this reduce the number of times required to rig up and lay down well test lines during transitions between phases but also gave us the capability to monitor the three wells simultaneously and identify any connectivity during fracs. More importantly, this being SDO's first attempt at Green Completions, the team needed ample time for the test package to be properly hooked up to the pipeline and clarify any interface ambiguity. The expected reduction in flaring is 90% compared to our previous development wells. This translates to 30,000+ tCO₂e flaring avoidance and an expected \$1 mmUSD hydrocarbon value per pad.

Finally, a robust Simultaneous Operations (SIMOPS) protocol was developed to govern operations on the pad. It involved Wells, Facilities, Civils and Service providers. The objective was to ensure that no SIMOPS are carried out in unsafe conditions and to set up a clear framework of accountability when they're allowed to occur.

The close proximity of the wells to each other gave the team a fast-tracked learning curve where improvements and efficiencies are captured and implemented immediately. Furthermore, downhole Formation behavior during Frac placement and cleanout influenced the design of the subsequent well on the pad ensuring a more robust execution.

Upon completion of the scope the team managed to achieve the following synergies and identify improvements for the next iteration:

- Consolidate data acquisition at pad level. It was agreed that performing 1 PLT instead of 3 for the pad would provide sufficient data for reservoir performance in that area of the field.
- Install lower assembly of frac tree at the end of drilling phase to reduce frac tree installation time at start of CWI campaign.
- Completed yard test for single lift using specially sourced 200T crane. This allowed us to reach all three wells on the pad whilst hoisting the Coil tubing or Wireline PCE, achieving a 40% reduction in BOP rig down and rig up for each well.
- Perform Cement Bond Log (CBL) logging on all 3 wells before perforation to reduce risk of wellbore exposure while waiting to frac, as cement logging is ongoing on other two wells. Further optimization was identified to allow hopping from one well to next without laying down BHA.
- One day of maintenance was built into the schedule. This eased the pressure on the team not to cut corners and really invest time on equipment maintenance to deliver the campaign. It also allowed the Shell team to manage demand for water and diesel delivery to support the operation.
- Recent risk of depletion identified the need to manage losses during Coil Tubing Clean Out (CTCO) with nitrified fluid an approach implemented for the first time in SDO operations.
- Planned green completion well testing (Flow to host) is expected to reduce GHG emissions from 30 to 1 ktCO₂e (90% reduction) and realize \$1 mmUSD in hydrocarbon value.

Overall the Frac and Test operation of the first Pad (Three wells) in Block 10 showed a 27.78% reduction in days compared to single wells Frac and Test period (Fig.8).

Looking ahead, the team will analyze the performance of the operations and assess if the extra investment in equipment to allow 'zipper frac' is justified. Also, we will target 0% flaring and strive for 100% GHG reduction in green completions.

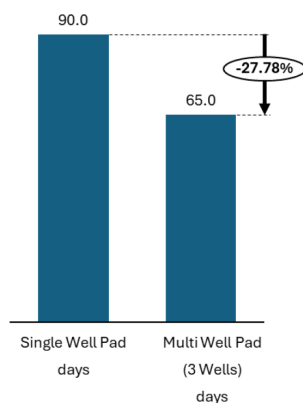


Figure 8—Frac and Test days of Single vs. Multi-well pads

Operations and Maintenance

Multi-well pad development offers significant operational and economic advantages over single-well configurations. By reducing the number of well pads by over 60%, multi-well pads contribute to capital expenditure (CAPEX) savings in the range of 15–20%. These savings are primarily driven by the reduction in flowlines, access roads, and utility installations such as power and communication lines. Unlike single-well pads, where each well typically requires a dedicated flowline to the manifold station, multi-well pads enable the use of shared trunklines or headers, reducing the total number of flowlines by up to a factor of three. Shorter distances between wells and the remote manifold station (RMS) further enhance cost efficiency.

The use of fewer flowlines simplifies pigging operations and reduces the complexity and cost of inspection, repair, and integrity management throughout the asset's lifecycle. As illustrated in Figure 8, flowline-related savings can reach approximately 50–60%. Additional cost reductions are observed in road construction (55–65%), electrical grid power installation (50–60%), and well pad development (20–25%), while savings from additional pipelines are considered negligible. Operational efficiency is also improved, as multiple wells can be monitored and managed from a single location, significantly reducing field time and enhancing overall productivity.

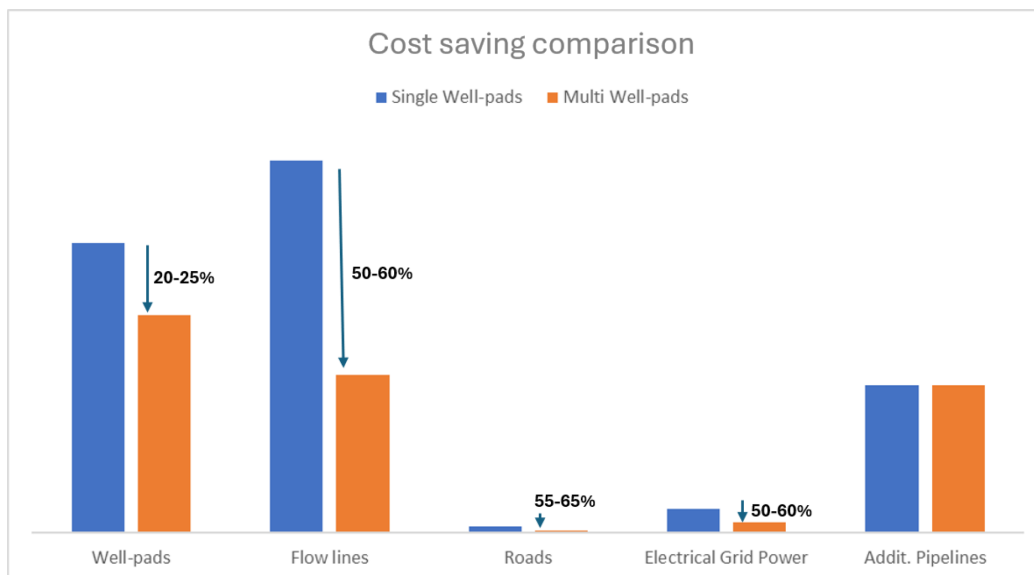


Figure 9—Savings from different aspects of multi-well pads vs. single pads

Discussion

The implementation of multi-well pad drilling in Block 10 has demonstrated a significant advancement in the development of deep tight gas reservoirs in Oman. The approach delivered measurable improvements in drilling efficiency, cost reduction, and environmental performance. Compared to traditional single-well development, the batch drilling of S-shaped deviated wells from triple-well pads enabled the reuse of BHAs, mud systems, and drilling roadmaps. These efficiencies translated into lower non-productive time and enhanced operational consistency across the campaign.

The integrated well maturation process—executed in clusters rather than individually—streamlined planning and reduced the timeline by approximately one-third. This was made possible by leveraging geological and operational similarities among wells, as well as adopting a consolidated technical assurance workflow. The success of this approach aligns with global best practices and validates the transferability of multi-well pad strategies from other Shell assets to the Omani context.

Completion and testing operations also benefited from the pad-based model. The use of a stack-frac approach with an Integrated Frac Services fleet allowed for shared mobilization and minimized equipment downtime. The introduction of green completions, with up to 90% reduction in GHG emissions, marked a significant step toward more sustainable operations. Additionally, the ability to monitor and test multiple wells from a single setup improved data acquisition efficiency and reduced surface intervention requirements.

While the campaign achieved its technical and safety objectives, challenges such as managing wellbore stability in reactive formations and optimizing directional profiles remain areas for future improvement. These learnings will inform the design and execution of subsequent pads, with a focus on refining well trajectories.

Overall, the results from this first multi-well pad campaign in Block 10 confirm the technical viability and scalability of the approach. The demonstrated benefits in efficiency, cost, and environmental impact provide a strong foundation for broader adoption across similar tight gas developments in Oman and the wider region.

Conclusion

The successful execution of multi-well pad drilling in Block 10 marks a significant milestone in Oman's tight gas development. By integrating pad-based planning, batch drilling, and shared infrastructure, the project achieved substantial reductions in well delivery time, rig mobilization, and operational costs. The use of S-shaped deviated wells, walking rigs, and standardized designs enabled efficient execution while maintaining high safety and environmental standards. Completion and testing operations further benefited from a stack-frac approach and green completions, resulting in up to 90% reduction in greenhouse gas emissions. The streamlined maturation process and collaborative workflows demonstrated the value of global knowledge sharing and cross-functional integration. This first-of-its-kind campaign sets a new performance baseline for tight gas development in Oman and provides a scalable, repeatable model for future phases and similar assets across the region.

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